

Computing Treewidth and Minimum Fill-In: All You Need are the Minimal Separators*

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Abstract. Consider a class of graphs $\mathcal{G}$ having a polynomial time algorithm computing the set of all minimal separators for every graph in $\mathcal{G}$. We show that there is a polynomial time algorithm for TREEWIDTH and MINIMUM FILL-IN, respectively, when restricted to the class $\mathcal{G}$. Many interesting classes of intersection graphs have a polynomial time algorithm computing all minimal separators, like permutation graphs, circle graphs, circular arc graphs, distance hereditary graphs, chordal bipartite graphs etc. Our result generalizes earlier results for the TREEWIDTH and MINIMUM FILL-IN for several of these classes. We also consider the related problems PATHWIDTH and INTERVAL COMPLETION when restricted to some special graph classes.

1 Introduction

In many recent investigations in computer science, the notions of treewidth and pathwidth play an increasingly important role. One reason for this is that many problems, including many well studied NP-complete graph problems, become solvable in polynomial and usually even linear time, when restricted to the class of graphs with bounded tree- or pathwidth [2, 4, 7]. Of crucial importance for these algorithms is, that a tree-decomposition or path-decomposition of the graph is given in advance. Much research has been done in finding a tree-decomposition with a reasonable small treewidth. Recently, it was shown in [8] that a linear time algorithm exists to find an optimal tree-decomposition for a graph with bounded treewidth. However, the constant hidden in the ‘big oh’, is exponential in the treewidth, limiting the practicality of this algorithm.

The problem ‘Given a graph $G = (V, E)$ and an integer k , is the treewidth of G at most k ’ is NP-complete, even when only complements of bipartite graphs G are allowed as input graphs [3] and it also remains NP-complete on bipartite graphs [20]. The problem ‘Given a graph $G = (V, E)$ and an integer k , is the pathwidth of G at most k ’ is NP-complete on cobipartite graphs [3], bipartite graphs [20] and triangulated graphs [17].

The treewidth can be computed in polynomial time on triangulated graphs (trivially), cographs [10], circular arc graphs [28], chordal bipartite graphs [23], permutation graphs [9], circle graphs [22] and distance hereditary graphs [1]. Since many NP-complete problems remain NP-complete when restricted to some of these classes, it is

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of great importance to be able to use the algorithms for graphs of small treewidth for these problems.

The TREewidth problem is the problem of finding a triangulated graph H with smallest maximum clique size having the given graph G as spanning subgraph. The PATHwidth problem is the problem of finding an interval graph H with smallest maximum clique size having the given graph G as spanning subgraph.

The following two problems have similar goals. The MINIMUM FILL-IN problem is the problem of finding a triangulated graph H having the given graph G as spanning subgraph such that the number of added edges is minimum. The INTERVAL COMPLETION problem is the problem of finding an interval graph H having the given graph G as spanning subgraph such that the number of added edges is minimum. The MINIMUM FILL-IN problem is also called CHORDAL GRAPH COMPLETION and is of great importance in relation with the performance of Gaussian elimination on sparse matrices, see [26].

The problem 'Given a graph $G = (V, E)$ and an integer k , is it possible to add at most k edges to G such that the new graph is triangulated', is NP-complete on cobipartite graphs [30] and on bipartite graphs [29]. On the other hand, in [27] an $O(n^5)$ algorithm for bipartite permutation graphs, a proper subclass of the chordal bipartite graphs, is given. Furthermore, a minimum fill-in can be computed in linear time on cographs [12], a proper subclass of the permutation graphs. The problem 'Given a graph $G = (V, E)$ and an integer k , is it possible to add at most k edges to G such that the new graph is an interval graph', is known to be NP-complete on line graphs [14].

It was shown in [9] that for every cocomparability graph G , i.e. G is a complement of a transitive orientable graph, the triangulated graph H having the graph G as a spanning subgraph and smallest maximum clique size is indeed an interval graph. Thus, treewidth and pathwidth coincide on cocomparability graph, a class of the perfect graphs containing permutation graphs, interval graphs and trapezoid graphs. R. Möhring [18] showed that this result is extendable to AT-free graphs (asteroidal triple-free graphs). Thus, on AT-free graphs, which contain cocomparability graphs as a proper subclass while they are no longer a subclass of perfect graphs, treewidth and pathwidth still coincide [24].

Moreover, the 'phase method', described in [18] yields a very general result which also applies to finding a triangulated graph H having the graph G as a spanning subgraph such that the number of added edges is minimum. Hence, if the graph G is AT-free, then such a graph H is always an interval graph. Consequently, the problems MINIMUM FILL-IN and INTERVAL COMPLETION coincide on AT-free graphs as did TREewidth and PATHwidth [24].

The main subject of the paper is to extend significantly the knowledge about the algorithmic complexity of the four mentioned problems when restricted to some special class of graphs by the following general approach:

Given a polynomial $P(n)$. Let $\mathcal{G}_P$ be a class of graphs having an algorithm computing for every $G \in \mathcal{G}_P$ the set of all minimal separators in $P(n)$ time, where n is the number of vertices of G . In this paper we show that for all graphs in such a class $\mathcal{G}_P$, the TREewidth and the MINIMUM FILL-IN can be computed in polynomial time. Notice that many interesting classes of graphs have a polynomial time algorithm computing the set of all minimal separators on all graphs of the class. Such classes are permutation graphs, circular permutation graphs, trapezoid graphs, circle graphs, circular arc graphs, distance hereditary graphs, chordal bipartite graphs, cocomparability graphs of bounded dimension, weakly triangulated graphs, etc. Our results generalize